Bird's-Eye-View Mapping from Vehicle Cameras via 3D Foundation Models

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I. INTRODUCTION

The ability to navigate complex environments safely requires the seamless execution of two critical, interconnected tasks: transforming raw sensor observations into a generalized understanding of the surroundings, and constructing maps that accurately position the agent relative to both static infrastructure and dynamic participants. In recent years, a unified Bird's-Eye-View (BEV) representation has emerged as the preferred medium for this purpose (Li et al., 2023; Ma et al., 2024). By uniting multiple sensing modalities into a single top-down perspective, BEV provides a highly informative representation that simplifies downstream navigation tasks such as motion planning and control, and aligns well with existing maps.

For ground-level users such as autonomous vehicles, bicycles, and pedestrians, attaining such a representation is non-trivial because primary sensors typically collect observations in perspective view (PV). Mapping from PV to BEV introduces several persistent challenges related to scale and resolution, geometric distortion, occlusions, and limited viewpoints. Historically, efforts to resolve these issues focused on bottom-up methods, including homography-based projections, explicit depth estimation, and Multi-Layer Perceptron (MLP) approaches (Ma et al., 2024). However, with the advent of the Transformer architecture which has seen increasing adoption in urban navigation research (Zheng et al., 2024; Li et al., 2024c; Hu et al., 2023), the field of BEV mapping has also shifted toward top-down transformer-based models such as BEVFormer (Li et al., 2024b). BEVFormer pioneered this trend by using learnable queries and spatiotemporal attention to search for features across views, a strategy that has since been refined by numerous successful successors.

Despite these advancements, most state-of-the-art methods rely on purpose-built, highly customized sensor configurations, typically vehicles equipped with rigid suites of six or more synchronized cameras. This reliance raises a critical question: how well do BEV methods hold up when restricted to more limited, mass-market inputs, such as standard dashcam imagery? While autonomous driving and dashcam-based accident forensics have historically been separate fields, a significant opportunity exists in the middle ground for proactively mapping road information and providing situational awareness without the strict safety constraints of active vehicle control. The rapidly evolving landscape and increasing heterogeneity of automotive cameras, illustrated in Figure 1, strongly motivate the development of robust perception frameworks capable of generalizing across mass-market imaging platforms.

In this work, we investigate whether extensive, specialized sensor suites are strictly necessary for these emerging applications. To address the reliance of standard BEV methods on precisely known extrinsics, we propose a novel augmented framework for BEV mapping that utilizes transformer-based BEV architectures alongside pre-trained 3D vision foundation models for metric camera pose estimation. We also conduct a comprehensive evaluation and analysis of the sensitivity of BEV methods to camera inputs. By systematically reducing the number of available cameras and perturbing input parameters, we evaluate the robustness of these models and demonstrate the feasibility of generating accurate BEV maps using hardware closer to mass-market devices than traditional research platforms. With this work, we aim to understand whether high-fidelity BEV mapping of road information is achievable without the extensive sensor suites typical of research vehicles, bridging the gap between high-end autonomous perception and consumer-grade road awareness applications.

Key Contributions

Our key contributions are listed as follows:

1. We propose a novel framework that replaces traditional, rigid camera calibration with metric poses regressed by a 3D foundation model, enabling BEV mapping from mass-market sensor inputs.
2. We experimentally validate our proposed framework combining the transformer-based BEV method, BEVFormer, and the 3D vision foundation model, MapAnything, on real-world data provided by the nuScenes dataset.
3. We systematically study the robustness of transformer-based BEV methods to changes in camera parameters such as the number of cameras, noise/occlusions, and ego poses/extrinsics.
4. We evaluate our framework to measure the sensitivity of foundation model-driven BEV mapping to degraded camera inputs.

Figure 1: The diverse and rapidly evolving landscape of automotive cameras. From left to right: (1) Google Dashcam feature on a Google Pixel 7 Pro phone mounted similarly to phones in the Google Smartphone Decimeter Challenge (Fu et al., 2022), image courtesy of 9to5Google; (2) popular dashcams of 2026 including (clockwise from top left) Redtiger, Rove, Rexing, Nextbase, and Viofo, image courtesy of Consumer Reports; (3) Gentex "Full Display Mirror" (FDM), image courtesy of Display Daily; (4) Mercedes-Benz Actros truck "MirrorCam" system, images courtesy of Daimler Truck AG; (5) "MirrorEye" system by Stoneridge, images courtesy of Stoneridge, Inc. and Lusilectra; and (6) roof-mounted sensor suite of a 5th-generation design Waymo, image courtesy of Waymo. Automotive cameras are being increasingly used for urban positioning and navigation, both on their own (Gupta and Gao, 2021) and in conjunction with other sensors (Mohanty et al., 2021; Bhamidipati and Gao, 2022; Wen et al., 2023; Zheng et al., 2023; Bai and Hsu, 2024).

II. APPROACH

Our work focuses on the challenge of processing PV imagery to generate BEV map representations which can subsequently be used for downstream perception tasks such as object detection. In designing our framework, we considered two broad classes of methods. The first constitutes transformer-based BEV models, which provide us with the desired BEV features as outputs but require known camera poses for the input images. The second method class constitutes 3D vision foundation models, which estimate camera poses as a byproduct of 3D scene reconstruction but do not directly produce BEV features as outputs. To leverage the strengths of both approaches, we thus propose the framework illustrated in Figure 2.

1. Transformer-Based BEV Models

Generating orthogonal BEV maps from PV images is challenging due to distance-based scale variance, the lack of explicit depth in monocular views, and foreground occlusions. Modern approaches to this mapping are categorized as bottom-up or top-down (Ma et al., 2024). Bottom-up methods correspond to solutions which lift 2D points into 3D space, while top-down methods correspond to solutions which project 3D targets back into 2D. The bottom-up methods can be further subdivided into three categories: homograph-based (Mallot et al., 1991), depth-based (Wang et al., 2019; Philion and Fidler, 2020), and MLP-based (Lu et al., 2019; Pan et al., 2020; Li et al., 2022).

Recently, the field has pivoted toward top-down, transformer-based architectures that sample features from PV images via learnable BEV queries. BEVFormer (Li et al., 2024b), the seminal architecture in this domain, extracts multi-scale 2D features from a six-camera rig and employs two core mechanisms:

- **Spatial Cross-Attention:** Lifts grid queries into 3D pillars and projects them onto 2D image planes using camera parameters to sample relevant pixels via deformable attention.
- **Temporal Self-Attention:** Fuses features from previous BEV frames with current observations, enabling the model to track velocity and maintain representations of temporarily occluded objects.

The success of BEVFormer on autonomous driving benchmarks has inspired numerous follow-up methods like SparseBEV (Liu et al., 2023), BEVNeXt (Li et al., 2024a), BEVSegFormer (Peng et al., 2023), and BEVMap (Chang et al., 2024). However, in this work we use the original BEVFormer method and its follow-on method, BEVFormer v2 (Yang et al., 2023), which have continued to enjoy widespread adoption amongst the autonomous driving and vision-centric BEV perception communities.

2. 3D Mapping and Vision Foundation Models

While BEV methods are the standard for transforming camera images into spatial features, most require precise camera intrinsics and extrinsics. As shown in Figure 1, consumer-grade setups often lack these parameters, necessitating 3D mapping approaches that can estimate poses from uncalibrated inputs.

Historically, 3D mapping transitioned from explicit geometric reconstruction to data-driven representations. Traditional Structure-from-Motion and SLAM frameworks such as COLMAP (Schonberger and Frahm, 2016) and ORB-SLAM (Mur-Artal et al., 2015) utilize multi-view geometry to triangulate coordinates. However, these methods often struggle with textureless surfaces or lighting extremes. The emergence of transformer-based architectures, such as the Visual Graph Grounded Transformer (Wang et al., 2025), marked a shift from rigid priors to learnable spatial relationships. Using cross-attention mechanisms, these models aggregate features across viewpoints to implicitly find correspondences without dense depth estimation.

This evolution has culminated in the current generation of 3D Foundation Models, with MapAnything (Keetha et al., 2025) representing the state of the art. Unlike specialized tools, MapAnything is a universal backbone that treats 3D reconstruction as a large-scale regression problem, allowing it to generalize across heterogeneous and uncalibrated sensors. Critically for our framework, MapAnything provides metric-scale distances and state-of-the-art pose estimation, making it the ideal foundation for our proposed pipeline.

3. Proposed Framework

In this work, we propose a framework that leverages the metric spatial awareness of 3D foundation models to enhance BEV synthesis. As illustrated in Figure 2, the pipeline begins with a vehicle trajectory that generates multi-view camera images. Rather than relying solely on potentially noisy or unavailable offline calibration, these images are processed by a 3D Foundation Model, such as MapAnything (Keetha et al., 2025), to regress precise, metric camera poses. These estimated poses are then fed alongside the original images into a transformer-based BEV model like BEVFormer (Li et al., 2024b). Within this architecture, the foundation model-derived poses serve as the geometric anchor for the spatial cross-attention mechanism, allowing the model to accurately project PV features into a unified BEV representation. By utilizing a foundation model as a robust pose engine, the framework aims to mitigate the performance degradation typically associated with calibration errors, yielding a more resilient and geographically consistent BEV Map.

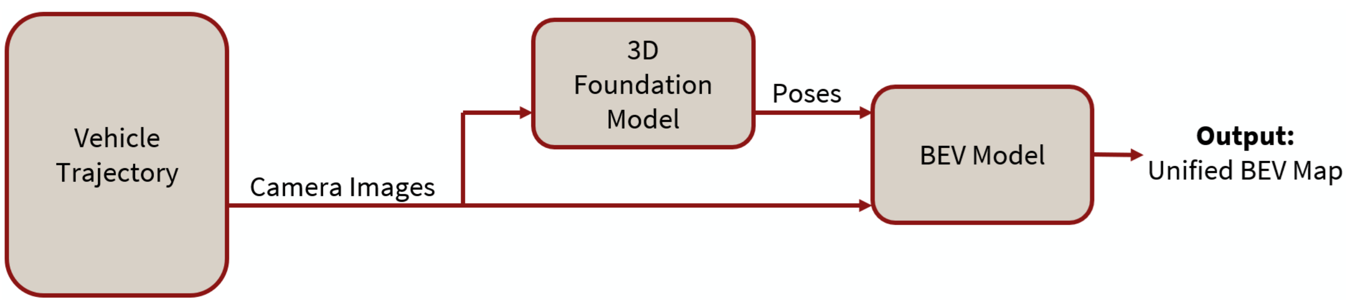

Figure 2: Our proposed framework to leverage the metric spatial awareness of 3D foundation models in conjunction with the powerful attention mechanisms of transformer-based BEV models for enhanced BEV synthesis.

III. EXPERIMENTS

In this section, we first detail our choice of dataset for evaluating our proposed framework. We next outline a comparative study between traditional ground-truth calibration and our proposed pipeline across multiple architectural scales of BEVFormer. By testing across these diverse conditions, we aim to quantify how effectively 3D foundation models can substitute for explicit camera parameters in unified BEV mapping.

1. nuScenes: A Real-World Dataset

While several high-quality datasets exist for urban navigation and autonomous driving research, such as UrbanNav (Hsu et al., 2023), KITTI (Geiger et al., 2013), and the Waymo Open Dataset (Sun et al., 2020), we have selected nuScenes (Caesar et al., 2020) as our primary testbed. This choice is driven by its status as the industry standard for BEV benchmarking, offering a comprehensive suite of sensor data including 360° camera coverage and high-quality 3D annotations of scene objects including velocity and attribute information. The dataset comprises 1,000 driving scenes recorded across Boston and Singapore, providing exceptional scene and object diversity across complex urban environments. These features make nuScenes particularly well-suited for evaluating the spatiotemporal dependencies and geometric robustness of modern transformer-based BEV models.

Figure 3 illustrates the nuScenes dataset at different scales: all scenes for a single location, individual ego vehicle poses for a single scene, and the time-synchronized camera frames captured from a specific pose.

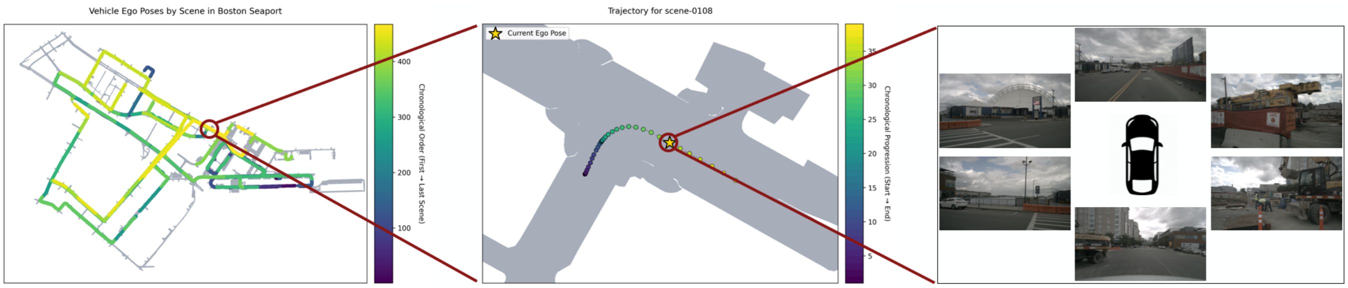

Figure 3: The nuScenes dataset (Caesar et al., 2020) illustrated at various scales. The left plot visualizes all 467 scenes from the Boston Seaport location color-coded by scene index. The center plot shows the individual ego poses for a single scene at that location, color-coded by timestamp. The scene follows a vehicle driving through a construction zone as it waits for another vehicle to pass and then turns right. The right plot shows the six raw camera images captured at a specific ego pose within the scene.

2. Experiments to Evaluate the Proposed Framework

To evaluate the efficacy of our proposed framework, we conduct experiments to compare our foundation model-driven approach with traditional calibration-reliant perception. The primary baseline for our study utilizes the standard BEVFormer architecture operating under nominal conditions with ground-truth camera parameters obtained from the nuScenes dataset. We then compare this baseline against our proposed framework, which substitutes these ground-truth poses with estimated camera extrinsics regressed by the MapAnything 3D foundation model. Figure 4 illustrates different views of the same scene from Figure 3 reconstructed in 3D using MapAnything. By keeping the raw camera images identical across both setups, we can directly assess how foundation model-derived poses influence the metrics outlined in the following section.

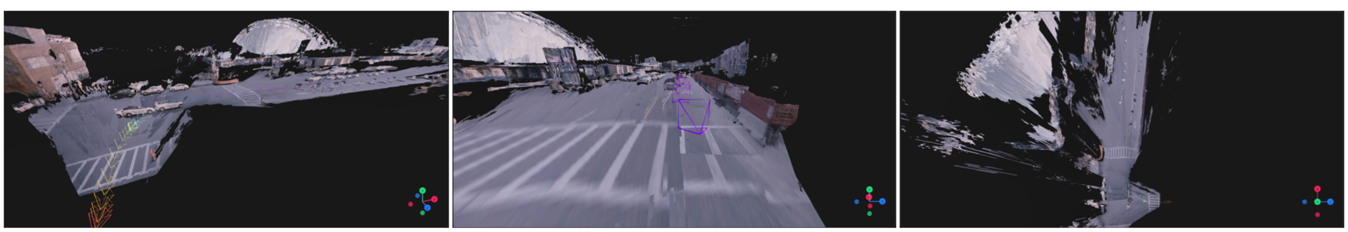

Figure 4: Different views of the same Boston Seaport scene from the nuScenes dataset (Caesar et al., 2020) in Figure 3, reconstructed in 3D using MapAnything (Keetha et al., 2025). The reconstruction was generated using only 39 images, all from the front camera alone. The center image looks similar to the perspective view seen in the raw front camera image in Figure 3. The right image showcases the natural and implicit ability of 3D foundation models like MapAnything to capture features in BEV even from a limited number of limited-perspective images.

To ensure our findings are generalizable across different architectural scales and versions, we conduct these evaluations across three distinct variants of the BEVFormer family: BEVFormer-Small, BEVFormer-Base, and BEVFormer-v2. Comparing these variants will reveal whether larger models with more complex attention mechanisms exhibit greater resilience to pose noise or if foundation models provide a consistent performance floor regardless of the downstream model's scale.

IV. EVALUATION METRICS

To evaluate the effectiveness of the proposed framework and the stability of the resulting representations, we employ both quantitative benchmarks and qualitative diagnostic analyses. This approach measures end-to-end perception accuracy while enabling inspection of the underlying transformer mechanisms. By correlating downstream task performance with internal attention behavior and feature-level shifts, we provide a comprehensive view of how foundation-model-derived poses and camera variations influence the reliability of the unified BEV map.

1. Object Detection Performance

As accurate BEV mapping is essential for autonomous navigation, we evaluate the system's primary perception capabilities using the BEVFormer architecture's native detection head. We utilize the nuScenes evaluation suite to quantify performance across various object classes.

- **mean Average Precision (mAP):** Measures general detection accuracy and classification success.
- **nuScenes Detection Score (NDS):** Consolidates mAP with errors in translation, scale, orientation, and velocity.

2. Characterizing Attention Robustness

Since attention is the core mechanism of transformer-based BEV models, we inspect the spatial and temporal consistency of these weights to diagnose failure modes like "geometric blindness."

- **Spatial Cross-Attention:** We assess the 3D-to-2D mapping by visualizing attention weights on input images. Successful mapping shows weights clustering near ground-truth projections, while failures appear as scattered or static points. We further quantify this using center offset error, which is the distance between the peak weight and the correct pixel coordinate.
- **Temporal Self-Attention:** To evaluate how the model leverages history, we analyze the mean Attribute Velocity Error. High velocity error despite low position error suggests a failure in fusing temporal offsets. We also visualize weight continuity to ensure the attention flow follows ego-motion-compensated paths.

3. Comparison of Resultant BEV Maps

We directly compare BEV maps generated under nominal conditions against those produced with perturbed camera inputs to measure feature-level integrity.

- **Feature Similarity:** We use the Structural Similarity Index (Wang et al., 2004) to evaluate spatial topology and cosine similarity on flattened feature tensors to determine if high-dimensional latent information remains intact despite input noise.
- **Geometric Displacement:** We track spatial shifts using Average Translation Error, measuring the Euclidean distance between object centers in nominal versus perturbed maps. To assess stability over time, we calculate centroid variance to quantify the "jitter" of object centers across the vehicle's trajectory.

V. ANALYSIS OF SENSITIVITY TO CAMERA INPUTS

To understand the robustness of the BEVFormer architecture, we perform a comprehensive sensitivity analysis investigating how variations in input quality and sensor configuration impact environmental perception. For these experiments, the model is trained exclusively under nominal conditions, utilizing the full six-camera suite with ground-truth extrinsic and intrinsic parameters and no synthetic noise. The resulting "perfect" case serves as the baseline for all subsequent evaluations. We then systematically perturb the evaluation input data across the following dimensions to quantify performance degradation:

1. **Reduced Camera Configurations:** Given the diversity of consumer-grade automotive camera setups in both number and quality, we wish to characterize how performance degrades when fewer than six cameras are available. To do this, we evaluate two handling strategies:
   - Camera Masking: Providing zero-initialized tensors for the missing camera feeds to simulate complete sensor loss. An example of this is visualized in Figure 5.
   - Input Duplication: Replicating an existing camera feed into the missing slot to maintain the expected input dimension.
2. **Image Noise and Occlusion Robustness:** We evaluate the model's resilience by introducing varying degrees of noise across a range of camera feeds, from a single perturbed camera to a total system-wide degradation. Of particular interest is how the BEV features in masked areas change when parts of the image are obscured. This allows us to determine if the spatial cross-attention can effectively compensate for missing data using neighboring views or temporal history.
3. **Spatial and Pose Perturbations:** We distinguish between errors in global localization and local sensor mounting to identify which geometric factor is more critical for BEV synthesis:
   - Ego-Pose Sensitivity: Perturbing the vehicle's estimated position and orientation within the world coordinate system.
   - Camera Extrinsics: Modifying the spatial relationship (translation and rotation) between the camera and the vehicle's center. This specifically tests the robustness of the spatial cross-attention's 3D-to-2D mapping.

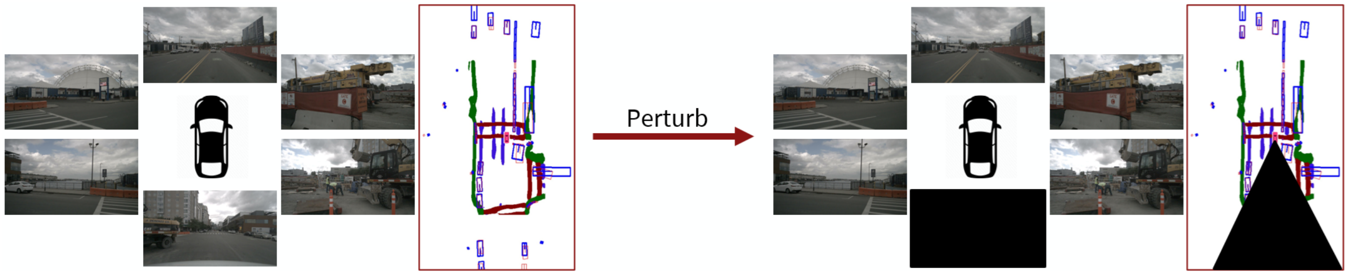

Figure 5: Visualization of the expected impact of a sample perturbation to the model's evaluation image inputs. Without any perturbations to the camera images, we would expect to obtain the complete BEV map (obtained here from BEVRestore (Kim et al., 2024)). With the back camera image masked out, however, we would expect a corresponding reduction in the generated BEV map as the rear part of the scene relative to the ego vehicle is no longer observable.

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